

COMPANION SLIDES

Chapter 8

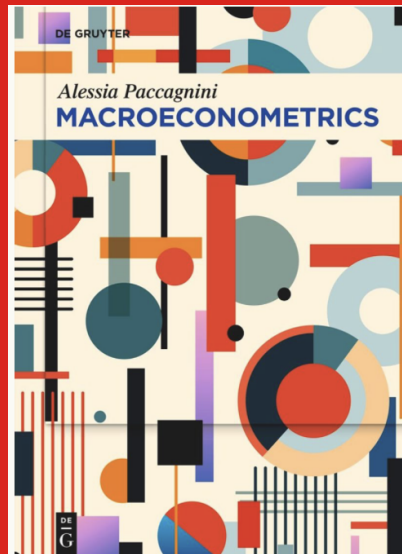
High Dimensionality

Extracting signal from large information sets

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Macroeconometrics – De Gruyter, 2026

DOI: 10.1515/9783111572802



The macroeconomic tension

Modern macroeconomists can observe **hundreds or thousands of indicators**, but the time dimension remains short.

- ▶ Central banks monitor large panels of real, nominal and financial variables.
- ▶ Digital sources add text, search data, social media and other alternative information.
- ▶ A VAR with n variables and p lags contains $n + n^2p$ coefficients.
- ▶ With $n = 20$ and $p = 4$: **1,620 parameters** – before adding richer dynamics.

How can we use more information without drowning in parameters?

Part I: Structured data

- ▶ Curse of dimensionality
- ▶ Principal components
- ▶ Static factor models
- ▶ Dynamic factor models
- ▶ Factor-Augmented VARs

Part II: Unstructured data

- ▶ Text as high-dimensional data
- ▶ Sentiment analysis
- ▶ Text factors and alternative data
- ▶ Topic models / LDA
- ▶ FAVAR empirical application

The unifying idea is **compression**: extract a small number of economically useful signals from a large information set.

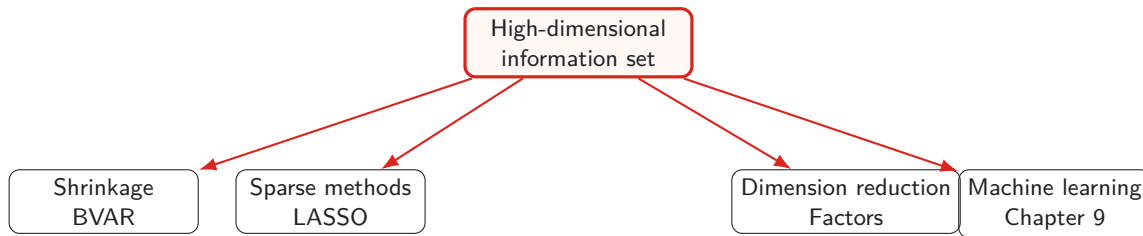
- ▶ Explain why conventional estimation deteriorates when dimension grows relative to sample size.
- ▶ Extract and interpret principal components from large panels.
- ▶ Distinguish exact and approximate factor models and understand rotational identification.
- ▶ Select the number of factors and understand why large n can improve factor estimation.
- ▶ Estimate dynamic factor models in state-space form and handle missing/mixed-frequency observations.
- ▶ Specify a FAVAR and recover impulse responses for many observables.
- ▶ Turn text into quantitative features and construct sentiment/text factors.
- ▶ Recognize the benefits and caveats of alternative data and topic models.

$$k = n + n^2 p$$

- ▶ Estimation variance rises as dimension increases.
- ▶ Overfitting can improve in-sample fit while harming forecasts.
- ▶ Highly correlated macro variables create multicollinearity.
- ▶ Classical approximations become unreliable when n/T is not small.

But many variables are valuable

- ▶ richer information sets
- ▶ robustness to variable choice
- ▶ cross-sectional information
- ▶ better measurement of latent concepts



Common logic: information is valuable, but unrestricted estimation is too expensive statistically.

Let $X_t = (X_{1t}, \dots, X_{nt})'$ be standardized. PCA finds orthogonal linear combinations with maximum variance.

$$F_{1t} = w_1' X_t, \quad \max_w \text{Var}(w' X_t) \quad \text{s.t. } w' w = 1$$

- ▶ w_1 is the eigenvector associated with the largest eigenvalue of Σ .
- ▶ Subsequent components maximize remaining variance subject to orthogonality.
- ▶ In practice, PCA can be computed efficiently by singular value decomposition.

Interpretation

A principal component is a **statistical summary**, not automatically an economically identified shock.

The fraction of variance explained by the first r components is

$$\frac{\sum_{j=1}^r \lambda_j}{\sum_{j=1}^n \lambda_j}$$

- ▶ Scree plots provide a first diagnostic.
- ▶ More components improve fit but can reintroduce noise.
- ▶ Factor-number selection should balance fit and parsimony.

Teaching idea

Ask students to run PCA on a macro panel and label the first few components using their largest loadings.

$$X_{it} = \lambda_i' F_t + e_{it}$$

- ▶ F_t : a small vector of common latent factors.
- ▶ λ_i : factor loadings linking variable i to the common component.
- ▶ e_{it} : idiosyncratic component.
- ▶ **Exact factor model**: strong restrictions on cross-sectional dependence in e_{it} .
- ▶ **Approximate factor model**: allows weak residual cross-correlation – more realistic for macro panels.

For any nonsingular matrix H ,

$$\Lambda F_t = (\Lambda H^{-1})(H F_t)$$

so factors and loadings are not uniquely identified without normalization.

What is identified?

The **space spanned by the factors** and the common component can be identified even though a particular rotation of the factors is not.

This matters when moving from *statistical factors* to *economic interpretation*.

- ▶ Bai–Ng information criteria penalize adding factors while rewarding improved fit.
- ▶ The optimal r minimizes an information criterion built from residual variance plus a dimension-dependent penalty.
- ▶ Under large- n , large- T asymptotics, factor estimates can converge at rate

$$\min(\sqrt{n}, \sqrt{T})$$

A high-dimensional surprise

Unlike unrestricted regression, adding informative series can **improve** estimation of the common factors.

Static factors summarize contemporaneous co-movement; dynamic factors model how common forces evolve over time.

$$X_t = \Lambda F_t + e_t, \quad F_t = A_1 F_{t-1} + \cdots + A_p F_{t-p} + u_t$$

- ▶ Dynamic structure captures persistence and propagation.
- ▶ State-space form opens the door to Kalman filtering and smoothing.
- ▶ Particularly useful with ragged edges, missing observations and mixed frequencies.

Measurement equation

$$X_t = \Lambda F_t + e_t$$

Observed indicators reveal the latent state imperfectly.

Transition equation

$$F_t = AF_{t-1} + u_t$$

The common factors follow their own dynamics.



- ▶ Dynamic common shocks can affect variables with different leads/lags.
- ▶ The generalized dynamic factor framework separates common and idiosyncratic spectral components.
- ▶ It is useful when contemporaneous PCA is too restrictive for the temporal structure of a large macro panel.

Practical message

Choose the factor architecture to match the information problem: static compression, dynamic forecasting, or structural analysis.

A small VAR may omit information available to policymakers. FAVAR augments the VAR with latent factors extracted from a large dataset.

$$\begin{bmatrix} F_t \\ Y_t \end{bmatrix} = \Phi(L) \begin{bmatrix} F_{t-1} \\ Y_{t-1} \end{bmatrix} + v_t$$

- ▶ F_t : latent information from the large panel.
- ▶ Y_t : observed policy variables such as the federal funds rate.
- ▶ After estimation, factor loadings map structural responses back to many observables.

Small VAR

A few variables must proxy for the information set of the central bank. Omitted information can distort identified shocks and generate puzzles.

FAVAR

Factors summarize a broad information set, while the VAR remains low-dimensional enough to estimate and interpret.

More information without estimating a VAR with hundreds of variables.

1. **Extract factors** from a large standardized macro panel using PCA.
2. **Estimate a VAR** on the factors and observed policy variables.
3. **Identify shocks** using the chosen structural restrictions.
4. **Project factor responses back** to individual observables through estimated loadings.

Two-step trade-off

Simple and transparent, but factor-estimation error can propagate into VAR estimates and recovered observable responses.

A document can contain thousands of potential features. The econometric problem is again one of compression.



- ▶ Tokenization, stopwords removal and stemming/lemmatization reduce noise.
- ▶ TF-IDF reweights terms by their informativeness across documents.

- ▶ A dictionary maps words or phrases to categories such as positive, negative, uncertainty or risk.
- ▶ Aggregate counts can be converted into time-series sentiment indices.
- ▶ The Economic Policy Uncertainty approach illustrates how text can proxy for latent macroeconomic concepts.

Trade-off

Dictionary methods are **transparent and reproducible**, but context, negation, domain shifts and language evolution can limit them.

- ▶ Apply PCA/factor methods to large sets of text-derived indicators.
- ▶ Combine textual and conventional macro data for forecasting and nowcasting.
- ▶ Google Trends, social media and other high-frequency sources can provide timely signals.
- ▶ The same high-dimensional caveats remain: selection bias, structural breaks, multiple testing and spurious correlation.

New data do not eliminate old econometric problems.

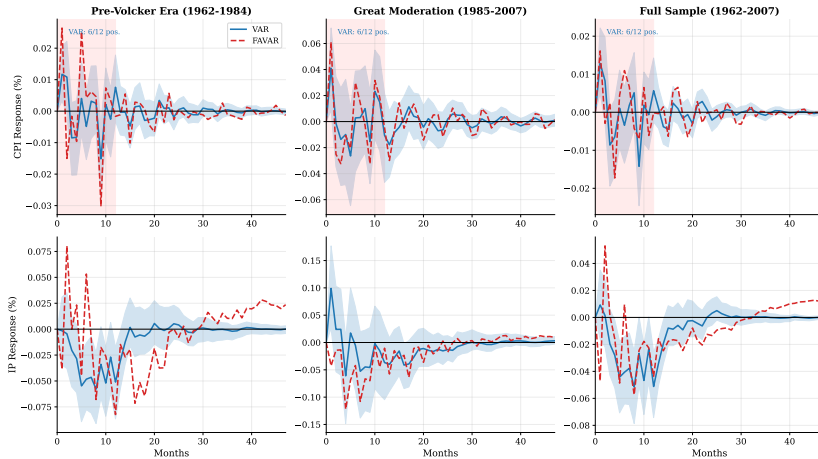
- ▶ Documents are mixtures of latent **topics**.
- ▶ Topics are probability distributions over words.
- ▶ Topic proportions provide low-dimensional document representations.
- ▶ LDA therefore has a natural analogy with factor models: latent components explain high-dimensional observables.

Applications

Central-bank communication, news narratives, policy themes and changing attention across time.

- ▶ Updated replication of Bernanke, Boivin and Elias using **FRED-MD**.
- ▶ Monthly panel: **126 series**, January 1962–December 2007.
- ▶ Benchmark VAR: industrial production growth, CPI inflation and federal funds rate; $p = 12$.
- ▶ FAVAR: extract **five principal components**, excluding the federal funds rate, then estimate a VAR on factors + FFR.
- ▶ Monetary policy shock: recursive identification, normalized to a **25 bp** increase in the federal funds rate.
- ▶ 90% bands from 500 bootstrap replications.

VAR vs FAVAR Across Monetary Policy Regimes **Response to 25bp Federal Funds Rate Increase**



- ▶ The three-variable VAR delivers plausible contractionary industrial-production responses.
- ▶ A **mild price puzzle** remains, but is attenuated relative to classic small-information-set exercises.
- ▶ The two-step FAVAR responses are noisier: factor estimation error propagates through the VAR and loading projection.
- ▶ Subsample patterns need not match simple intuition; regime changes alter both dynamics and the estimated factor space.
- ▶ Replication with updated data is informative precisely because results need not be textbook-perfect.

Rolling one-step-ahead FFR forecasts, initial window 120 months:

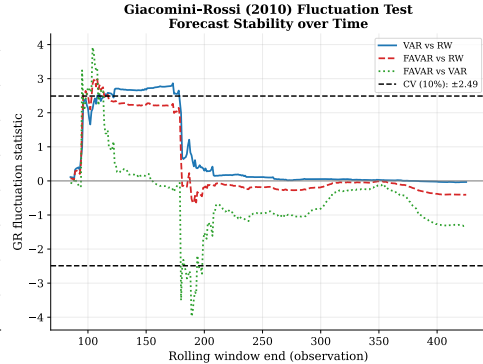
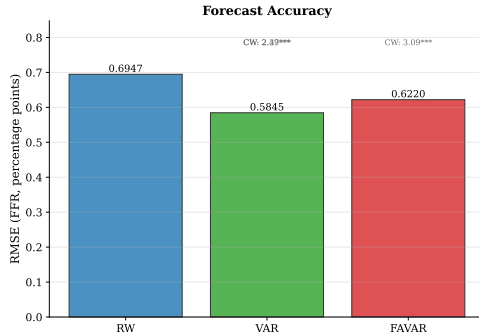
Model	RMSE	CW stat
RW	0.695	–
VAR	0.585	2.49***
FAVAR	0.622	3.09***

Result

VAR > FAVAR > RW in point-forecast accuracy for the federal funds rate in this exercise.

- ▶ Both richer models beat the RW significantly.
- ▶ Relative performance is not stable over time under the Giacomini–Rossi fluctuation test.

Out-of-Sample Forecast Comparison: FFR
RW vs VAR vs FAVAR | Clark-West & Giacomini-Rossi Tests



Goal	Method	Why
Compress a large panel	PCA	Simple, fast statistical summary
Estimate common macro forces	Factor model	Explicit common/idiosyncratic decomposition
Nowcast with ragged data	Dynamic factor model	State-space + Kalman filter
Structural analysis with rich information	FAVAR	Factors + VAR identification
Measure tone / uncertainty	Dictionary sentiment	Transparent text index
Discover themes	LDA / topic model	Latent low-dimensional topics

- ▶ **More data is not automatically better:** irrelevant or unstable series can add noise.
- ▶ PCA maximizes variance, not economic relevance or causal identification.
- ▶ Factor-number selection and data transformations matter.
- ▶ Two-step procedures understate uncertainty if factor estimation error is ignored.
- ▶ Text features are sensitive to preprocessing, vocabulary and context.
- ▶ Alternative data can suffer from selection bias and changing platform behavior.

1. High dimensionality is both an **opportunity** and a statistical constraint.
2. Factor methods compress large information sets into a tractable number of common signals.
3. Dynamic factors and FAVARs make compression useful for forecasting and structural analysis.
4. Text and alternative data extend the same logic to unstructured information.
5. High-dimensional methods require disciplined validation: transformations, factor choice, stability and genuine out-of-sample performance all matter.

- ▶ Run PCA on a macro panel and interpret the first three factors from their loadings.
- ▶ Compare Bai–Ng factor selection with a scree-plot rule.
- ▶ Estimate a dynamic factor model with an artificial ragged edge.
- ▶ Compare small-VAR and FAVAR monetary-policy IRFs under alternative factor counts.
- ▶ Construct a simple uncertainty/sentiment index from central-bank statements or news.
- ▶ Compare dictionary sentiment with a supervised text measure.
- ▶ Re-run the forecasting exercise over a different policy regime or target variable.

A flexible starting point

These companion slides summarize selected ideas from the chapter. They are intentionally editable rather than exhaustive.

- ▶ Add your own datasets, factor interpretations and FAVAR applications.
- ▶ Insert coding demonstrations in R, Python, MATLAB, Julia or Stata.
- ▶ Add current text/alternative-data examples relevant to your course.
- ▶ Expand derivations or simplify them depending on the audience.
- ▶ Add exercises, discussion questions and replication assignments.

Make them yours: add examples, applications, exercises and discussion questions.

From compression to flexible prediction

Chapter 8 asks how to summarize a large information set. Chapter 9 asks how to use regularization and nonlinear learning algorithms when the predictive relationship itself may be complex.

High-dimensional data $\longrightarrow$ Machine learning